
grokking
Deep Reinforcement
Learning
Miguel Morales

Grokking Deep Reinforcement Learning

MIGUEL MORALES

FOREWORD BY CHARLES ISBELL, JR.

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dedication

For Danelle, Aurora, Solomon, and those to come.

Being with you is a +1 per timestep.

(You can safely assume +1 is the highest reward.)

I love you!

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foreword

So, here's the thing about reinforcement learning. It is difficult to learn and difficult to teach, for a number of reasons. First, it's quite a technical topic. There is a great deal of math and theory behind it. Conveying the right amount of background without drowning in it is a challenge in and of itself.

Second, reinforcement learning encourages a conceptual error. RL is both a way of thinking about decision-making problems and a set of tools for solving those problem. By "a way of thinking," I mean that RL provides a framework for making decisions: it discusses states and reinforcement signals, among other details. When I say "a set of tools," I mean that when we discuss RL, we find ourselves using terms like *Markov decision processes* and *Bellman updates*. It is remarkably easy to confuse the way of thinking with the mathematical tools we use in response to that way of thinking.

Finally, RL is implementable in a wide variety of ways. Because RL is a way of thinking, we can discuss it by trying to realize the framework in a very abstract way, or ground it in code, or, for that matter, in neurons. The substrate one decides to use makes these two difficulties even more challenging—which bring us to deep reinforcement learning.

Focusing on deep reinforcement learning nicely compounds all these problems at once. There is background on RL, and background on deep neural networks. Both are separately worthy of study and have developed in completely different ways. Working out how to explain both in the context of developing tools is no easy task. Also, do not forget that understanding RL requires understanding not only the tools and their realization in deep networks, but also understanding the way of thinking about RL; otherwise, you cannot generalize beyond the examples you study directly. Again, teaching RL is hard, and there are so many ways for teaching deep RL to go wrong—which brings us to Miguel Morales and this book.

This book is very well put together. It explains in technical but clear language what machine learning is, what deep learning is, and what reinforcement learning is. It allows the reader to understand the larger context of where the field is and what you can do with the techniques of deep RL, but also the way of thinking that ML, RL, and deep RL present. It is clear and concise. Thus, it works as both a learning guide and as a reference, and, at least for me, as a source of some inspiration.

I am not surprised by any of this. I've known Miguel for quite a few years now. He went from taking machine learning courses to teaching them. He has been the lead teaching assistant on my Reinforcement Learning and Decision Making course for the Online Masters of Science at Georgia Tech for more semesters than I can count. He's reached thousands of students during that time. I've watched him grow as a practitioner, a researcher, and an educator. He has helped to make the RL course at GT better than it started out, and continues even as I write this to make the experience of grokking reinforcement learning a deeper one for the students. He is a natural teacher.

This text reflects his talent. I am happy to be able to work with him, and I'm happy he's been moved to write this book. Enjoy. I think you'll learn a lot. I learned a few things myself.

Charles Isbell, Jr.

Professor and John P. Imlay Jr. Dean

College of Computing

Georgia Institute of Technology

preface

Reinforcement learning is an exciting field with the potential to make a profound impact on the history of humankind. Several technologies have influenced the history of our world and changed the course of humankind, from fire, to the wheel, to electricity, to the internet. Each technological discovery propels the next discovery in a compounding way. Without electricity, the personal computer wouldn’t exist; without it, the internet wouldn’t exist; without it, search engines wouldn’t exist.

To me, the most exciting aspect of RL and artificial intelligence, in general, is not so much to merely have other intelligent entities next to us, which is pretty exciting, but instead, what comes after that. I believe reinforcement learning, being a robust framework for optimizing specific tasks autonomously, has the potential to change the world. In addition to task automation, the creation of intelligent machines may drive the understanding of human intelligence to places we have never been before. Arguably, if you can know with certainty how to find optimal decisions for every problem, you likely understand the algorithm that finds those optimal decisions. I have a feeling that by creating intelligent entities, humans can become more intelligent beings.

But we are far away from this point, and to fulfill these wild dreams, we need more minds at work. Reinforcement learning is not only in its infancy, but it’s been in that state for a while, so there is much work ahead. The reason I wrote this book is to get more people grokking deep RL, and RL in general, and to help you contribute.

Even though the RL framework is intuitive, most of the resources out there are difficult to understand for newcomers. My goal was not to write a book that provides code examples only, and most definitely not to create a resource that teaches the theory of reinforcement learning. Instead, my goal was to create a resource that can bridge the gap between theory and practice. As you’ll soon see, I don’t shy away from equations; they are essential if you want to grok a research field. And, even if your goal is practical, to build quality RL solutions, you still need that theoretical foundation. However, I also don’t solely rely on equations because not everybody interested in RL is fond of math. Some people are more comfortable with code and concrete examples, so this book provides the practical side of this fantastic field.

Most of my effort during this three-year project went into bridging this gap; I don’t shy away from intuitively explaining the theory, and I don’t just plop down code examples. I do both, and in a very detail-oriented fashion. Those who have a hard time understanding the textbooks and lectures can more easily grasp the words top researchers use: why those specific words, why not other words. And those who know the words and love reading the equations but have trouble seeing those equations in code and how they connect can more easily understand the practical side of reinforcement learning.

Finally, I hope you enjoy this work, and more importantly that it does fulfill its goal for you. I hope that you emerge grokking deep reinforcement learning and can give back and contribute to this fantastic community that I've grown to love. As I mentioned before, you wouldn't be reading this book if it wasn't for a myriad of relatively recent technological innovations, but what happens after this book is up to you, so go forth and make an impact in the world.

acknowledgments

I want to thank the people at Georgia Tech for taking the risk and making available the first Online Master of Science in Computer Science for anyone in the world to get a high-quality graduate education. If it weren't for those folks who made it possible, I probably would not have written this book.

I want to thank Professor and Dean Charles Isbell and Professor Michael Littman for putting together an excellent reinforcement-learning course. I have a special appreciation for Dean Isbell, who has given me much room to grow and learn RL. Also, the way I teach reinforcement learning—by splitting the problem into three types of feedback—I learned from Professor Littman. I'm grateful to have received instruction from them.

I want to thank the vibrant teaching staff at Georgia Tech's CS 7642 for working together on how to help students learn more and enjoy their time with us. Special thanks go to Tim Bail, Pushkar Kolhe, Chris Serrano, Farrukh Rahman, Vahe Hagopian, Quinn Lee, Taka Hasegawa, Tianhang Zhu, and Don Jacob. You guys are such great teammates. I also want to thank the folks who previously contributed significantly to that course. I've gotten a lot from our interactions: Alec Feuerstein, Valkyrie Felso, Adrien Ecoffet, Kaushik Subramanian, and Ashley Edwards. I want to also thank our students for asking the questions that helped me identify the gaps in knowledge for those trying to learn RL. I wrote this book with you in mind. A very special thank you goes out to that anonymous student who recommended me to Manning for writing this book; I still don't know who you are, but you know who you are. Thank you.

I want to thank the folks at Lockheed Martin for all their feedback and interactions during my time writing this book. Special thanks go to Chris Aasted, Julia Kwok, Taylor Lopez, and John Haddon. John was the first person to review my earliest draft, and his feedback helped me move the writing to the next level.

I want to thank the folks at Manning for providing the framework that made this book a reality. I thank Brian Sawyer for reaching out and opening the door; Bert Bates for setting the compass early on and helping me focus on teaching; Candace West for helping me go from zero to something; Susanna Kline for helping me pick up the pace when life got busy; Jennifer Stout for cheering me on through the finish line; Rebecca Rinehart for putting out fires; Al Krinker for providing me with actionable feedback and helping me separate the signal from the noise; Matko Hrvatin for keeping up with MEAP releases and putting that extra pressure on me to keep writing; Candace Gillhoolley for getting the book out there, Stjepan Jureković for getting me out there; Ivan Martinovic for getting the much-needed feedback to improve the text; Lori Weidert for aligning the book to be production-ready twice; Jennifer Houle for being gentle with the design changes; Katie Petito for patiently working through the details; Katie Tennant for the meticulous and final polishing touches; and to anyone I missed, or who worked behind the scenes to make this book a reality. There are more, I know: thank you all for your hard work.

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I want to thank the folks at Udacity for letting me share my passion for this field with their students and record the actor-critic lectures for their Deep Reinforcement Learning Nanodegree. Special thanks go to Alexis Cook, Mat Leonard, and Luis Serrano.

I want to thank the RL community for helping me clarify the text and improve my understanding. Special thanks go to David Silver, Sergey Levine, Hado van Hasselt, Pascal Poupart, John Schulman, Pieter Abbeel, Chelsea Finn, Vlad Mnih, for their lectures; Rich Sutton for providing the gold copy of the field in a single place (his textbook); and James MacGlashan, and Joshua Achiam for their codebases, online resources, and guidance when I didn't know where to go to get an answer to a question. I want to thank David Ha for giving me insights as to where to go next.

Special thanks go to Silvia Mora for helping make all the figures in this book presentable and helping me in almost every side project that I undertake.

Finally, I want to thank my family, who were my foundation throughout this project. I "knew" writing a book was a challenge, and then I learned. But my wife and kids were there regardless, waiting for my 15-minute breaks every 2 hours or so during the weekends. Thank you, Solo, for brightening up my life midway through this book. Thank you, Rosie, for sharing your love and beauty, and thank you Danelle, my wonderful wife, for everything you are and do. You are my perfect teammate in this interesting game called life. I'm so glad I found you.

about this book

Grokking Deep Reinforcement Learning bridges the gap between the theory and practice of deep reinforcement learning. The book's target audience is folks familiar with machine learning techniques, who want to learn reinforcement learning. The book begins with the foundations of deep reinforcement learning. It then provides an in-depth exploration of algorithms and techniques for deep reinforcement learning. Lastly, it provides a survey of advanced techniques with the potential for making an impact.

Who should read this book

Folks who are comfortable with a research field, Python code, a bit of math here and there, lots of intuitive explanations, and fun and concrete examples to drive the learning will enjoy this book. However, any person only familiar with Python can get a lot, given enough interest in learning. Even though basic DL knowledge is assumed, this book provides a brief refresher on neural networks, backpropagation, and related techniques. The bottom line is that this book is self contained, and anyone wanting to play around with AI agents and emerge grokking deep reinforcement learning can use this book to get them there.

How this book is organized: a roadmap

This book has 13 chapters divided into two parts.

In part 1, chapter 1 introduces the field of deep reinforcement learning and sets expectations for the journey ahead. Chapter 2 introduces a framework for designing problems that RL agents can understand. Chapter 3 contains details of algorithms for solving RL problems when the agent knows the dynamics of the world. Chapter 4 contains details of algorithms for solving simple RL problems when the agent does not know the dynamics of the world. Chapter 5 introduces methods for solving the prediction problem, which is a foundation for advanced RL methods.

In part 2, chapter 6 introduces methods for solving the control problem, methods that optimize policies purely from trial-and-error learning. Chapter 7 teaches more advanced methods for RL, including methods that use planning for more sample efficiency. Chapter 8 introduces the use of function approximation in RL by implementing a simple RL algorithm that uses neural networks for function approximation. Chapter 9 dives into more advanced techniques for using function approximation for solving reinforcement learning problems. Chapter 10 teaches some of the best techniques for further improving the methods introduced so far. Chapter 11 introduces a slightly different technique for using DL models with RL that has proven to reach state-of-the-art performance in multiple deep RL benchmarks. Chapter 12 dives into more advanced methods for deep RL, state-of-the-art algorithms, and techniques commonly used for solving real-world problems. Chapter 13 surveys advanced research areas in RL that suggest the best path for progress toward artificial general intelligence.

About the code

This book contains many examples of source code both in boxes titled "I speak Python" and in the text. Source code is formatted in a `fixed-width font like this` to separate it from ordinary text and has syntax highlighting to make it easier to read.

In many cases, the original source code has been reformatted; we've added line breaks, renamed variables, and reworked indentation to accommodate the available page space in the book. In rare cases, even this was not enough, and code includes line-continuation operator in Python, the backslash (\), to indicate that a statement is continued on the next line.

Additionally, comments in the source code have often been removed from the boxes, and the code is described in the text. Code annotations point out important concepts.

The code for the examples in this book is available for download from the Manning website at https://www.manning.com/books/grokking-deep-reinforcement-learning and from GitHub at https://github.com/mimoralea/gdrl.

liveBook discussion forum

Purchase of *Grokking Deep Reinforcement Learning* includes free access to a private web forum run by Manning Publications where you can make comments about the book, ask technical questions, and receive help from the author and from other users. To access the forum, go to https://livebook.manning.com/#!/book/grokking-deep-reinforcement-learning/discussion. You can also learn more about Manning's forums and the rules of conduct at https://livebook.manning.com/#!/discussion.

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about the author

MIGUEL MORALES works on reinforcement learning at Lockheed Martin, Missiles and Fire Control, Autonomous Systems, in Denver, Colorado. He is a part-time Instructional Associate at Georgia Institute of Technology for the course in Reinforcement Learning and Decision Making. Miguel has worked for Udacity as a machine learning project reviewer, a Self-driving Car Nanodegree mentor, and a Deep Reinforcement Learning Nanodegree content developer. He graduated from Georgia Tech with a Master's in Computer Science, specializing in interactive intelligence.

1 Introduction to deep reinforcement learning

In this chapter

- **You will learn what deep reinforcement learning is and how it is different from other machine learning approaches.**
- **You will learn about the recent progress in deep reinforcement learning and what it can do for a variety of problems.**

- **You will know what to expect from this book and how to get the most out of it.**

> ***I visualize a time when we will be to robots what dogs are to humans, and I'm rooting for the machines.***
>
> **— Claude Shannon**
> **Father of the information age and contributor to the field of artificial intelligence**

Humans naturally pursue feelings of happiness. From picking out our meals to advancing our careers, every action we choose is derived from our drive to experience rewarding moments in life. Whether these moments are self-centered pleasures or the more generous of goals, whether they bring us immediate gratification or long-term success, they're still our perception of how important and valuable they are. And to some extent, these moments are the reason for our existence.

Our ability to achieve these precious moments seems to be correlated with intelligence; "intelligence" is defined as the ability to acquire and apply knowledge and skills. People who are deemed by society as intelligent are capable of trading not only immediate satisfaction for long-term goals, but also a good, certain future for a possibly better, yet uncertain, one. Goals that take longer to materialize and that have unknown long-term value are usually the hardest to achieve, and those who can withstand the challenges along the way are the exception, the leaders, the intellectuals of society.

In this book, you learn about an approach, known as deep reinforcement learning, involved with creating computer programs that can achieve goals that require intelligence. In this chapter, I introduce deep reinforcement learning and give suggestions to get the most out of this book.

What is deep reinforcement learning?

Deep reinforcement learning (DRL) is a machine learning approach to artificial intelligence concerned with creating computer programs that can solve problems requiring intelligence. The distinct property of DRL programs is learning through trial and error from feedback that's simultaneously sequential, evaluative, and sampled by leveraging powerful non-linear function approximation.

I want to unpack this definition for you one bit at a time. But, don't get too caught up with the details because it'll take me the whole book to get you grokking deep reinforcement learning. The following is the introduction to what you learn about in this book. As such, it’s repeated and explained in detail in the chapters ahead.

If I succeed with my goal for this book, after you complete it, you should understand this definition precisely. You should be able to tell why I used the words that I used, and why I didn’t use more or fewer words. But, for this chapter, simply sit back and plow through it.

Deep reinforcement learning is a machine learning approach to artificial intelligence

Artificial intelligence (AI) is a branch of computer science involved in the creation of computer programs capable of demonstrating intelligence. Traditionally, any piece of software that displays cognitive abilities such as perception, search, planning, and learning is considered part of AI. Several examples of functionality produced by AI software are

- The pages returned by a search engine
- The route produced by a GPS app
- The voice recognition and the synthetic voice of smart-assistant software